



Lecture 21:

Retrieval-Augmented and Knowledge-Augmented Language Models

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Semester 1, 2026

*School of Computing and Information Systems
University of Melbourne*



Lecture 21:

Retrieval-Augmented and Knowledge-Augmented Language Models

1. Retrieval Augmented LMs

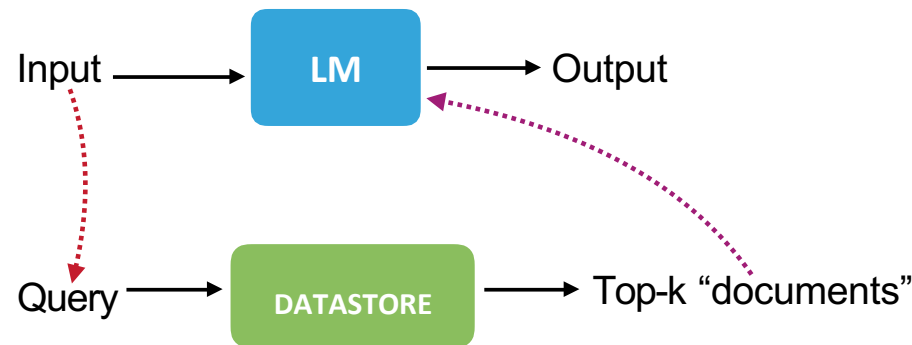
- What is Retrieval-Augmented Language Models
- Why Retrieval-Augmented LMs?
- RAG Framework and Training Methods

2. Knowledge with LLM

- How to Represent the Knowledge?
- Language Models with Knowledge Base?
- Language Models with Commonsense Knowledge Base?

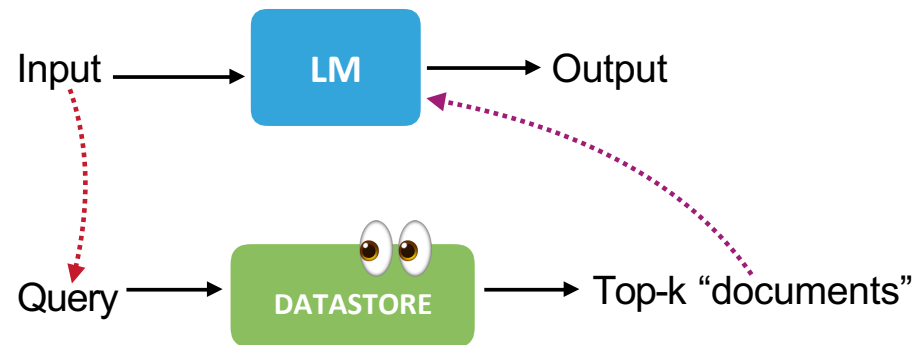
Retrieval-Augmented Language Models (RALMs)

- Language models that use an **external datastore** (or **knowledge base**) to improve their responses by retrieving relevant information during inference.
- *Instead of relying only on their internal knowledge*, these models "**look up**" supporting documents or facts from the **datastore** before generating an answer.



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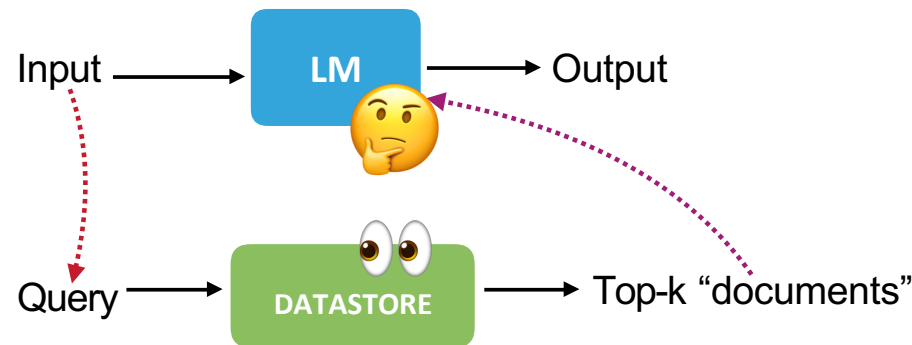


- **Datastore:**
 - What should be **stored** in the datastore?
 - How to **index** the datastore (e.g., granularity)?
 - How to **search** top-k documents efficiently?

1 What is Retrieval-Augmented LMs?

Retrieval-Augmented Language Models (RALMs)

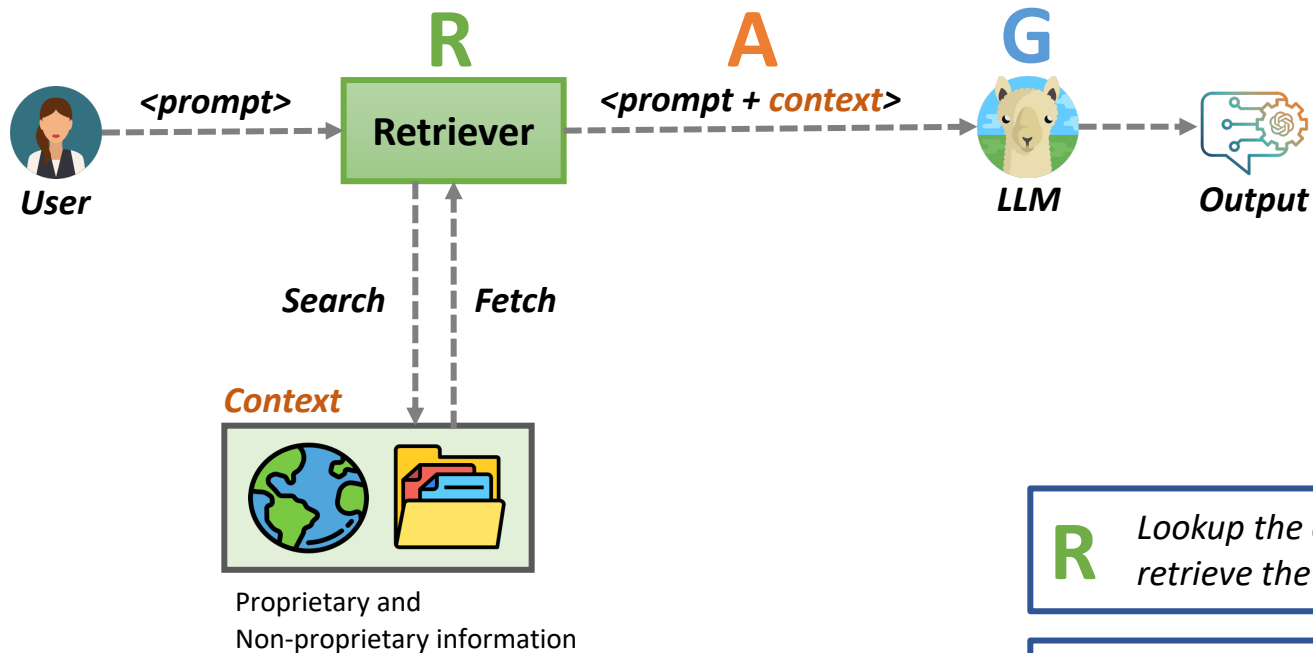
- Language models that use an **external datastore** (or **knowledge base**) to improve their responses by retrieving relevant information during inference.
- *Instead of relying only on their internal knowledge*, these models "**look up**" supporting documents or facts from the **datastore** before generating an answer.



- How to integrate the *retrieved sources (Top-k "documents")* with LMs?
- You can also search the **datastore** in multiple rounds.

1 What is Retrieval-Augmented LMs?

Retrieval Augmented Generation Framework!



R Lookup the external source to retrieve the relevant information

A Add the retrieved information to the user prompt

G Use LLM to generate response to user prompt with the context

Path #1: *Build a language model that has a built-in retrieval component*

- Consider **how to build the datastore/index** as part of the model
 - *e.g. use sentence/phrase-based storage*
 - *e.g. convert sentences as query vectors*
- An alternative of **scaling today's parametric** (Transformer-based) **LMs**
- Lots of **interesting technical challenges** (including retrieval performance, speed, integration methods), not as successful as we hoped for



Improving language models by retrieving from trillions of tokens

Sebastian Borgeaud[†], Arthur Mensch[†], Jordan Hoffmann[†], Trevor Cai, Eliza Rutherford, Katie Millican, George van den Driessche, Jean-Baptiste Lespiau, Bogdan Damoc, Aidan Clark, Diego de Las Casas, Aurelia Guy, Jacob Menick, Roman Ring, Tom Hennigan, Saffron Huang, Loren Maggiore, Chris Jones, Albin Cassirer, Andy Brock, Michela Paganini, Geoffrey Irving, Oriol Vinyals, Simon Osindero, Karen Simonyan, Jack W. Rae[‡], Erich Elsen[‡] and Laurent Sifre^{†,‡}

All authors from DeepMind, [†]Equal contributions, [‡]Equal senior authorship

RETRO (Borgeaud et al., 2022)

1 What is Retrieval-Augmented LMs?

Path #2: Consider retrieval as one of the tools that LMs learn how to use

Assuming you already have a **very powerful LM**

- Retrieval can be viewed as an **API** or “**black box**” (e.g., a search engine)
- When to call the retriever? how to take retrieved results in context?
- Very popular in developer community (“Frozen RAG”)
 - **Frozen LLM** (without any fine-tuning) + **Retrieval API**

The New England Journal of Medicine is a registered trademark of [QA(“Who is the publisher of The New England Journal of Medicine?”) → Massachusetts Medical Society] the MMS.

Out of 1400 participants, 400 (or [Calculator(400 / 1400) → 0.29] 29%) passed the test.

The name derives from “la tortuga”, the Spanish word for [MT(“tortuga”) → turtle] turtle.

The Brown Act is California’s law [WikiSearch(“Brown Act”) → The Ralph M. Brown Act is an act of the California State Legislature that guarantees the public’s right to attend and participate in meetings of local legislative bodies.] that requires legislative bodies, like city councils, to hold their meetings open to the public.

Figure 1: Exemplary predictions of Toolformer. The model autonomously decides to call different APIs (from top to bottom: a question answering system, a calculator, a machine translation system, and a Wikipedia search engine) to obtain information that is useful for completing a piece of text.

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Retrieval-Augmented and Knowledge-Augmented Language Models

1. Retrieval Augmented LMs

- What is Retrieval-Augmented Language Models
- **Why Retrieval-Augmented LMs?**
- RAG Framework and Training Methods

2. Knowledge with LLM

- How to Represent the Knowledge?
- Language Models with Knowledge Base?
- Language Models with Commonsense Knowledge Base?

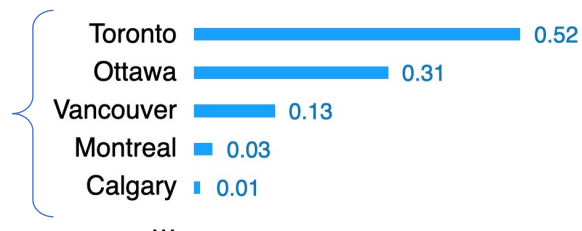
1 Why Retrieval-Augmented LMs?

Retrieval-augmented LMs

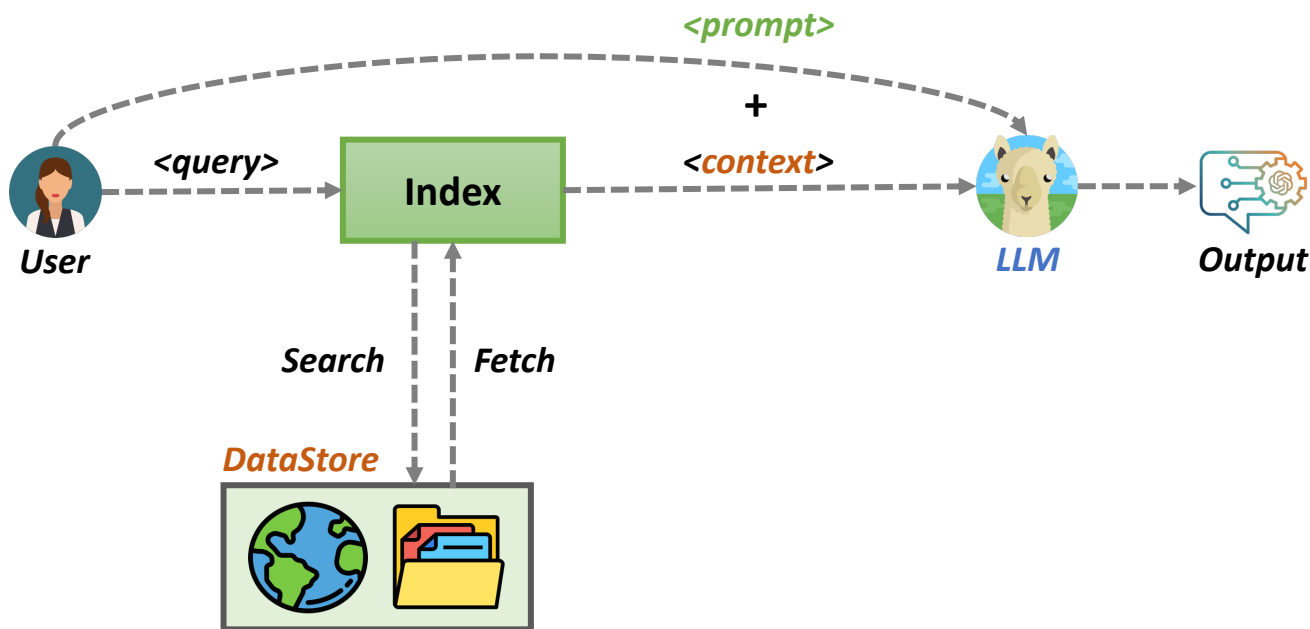
- It is a **language model**

$$P(x_n | x_1, x_2, \dots, x_{n-1})$$

The capital city of Ontario is ____



- It retrieves from an **external datastore** (at least during inference time)



1 Why Retrieval-Augmented LMs?

Can you cram all knowledge into parameters?

who is the president of United States?



As of January 2025, the President of the United States is Joe Biden. He was inaugurated for his second term on January 20, 2025, after winning the 2024 presidential election.



(Jan 23, 2025)



1 Why Retrieval-Augmented LMs?

Can you cram all knowledge into parameters?

what is the most cited published of yann lecun

The most cited publication by Yann LeCun is generally considered to be:

"Gradient-Based Learning Applied to Document Recognition" (1998)

by Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner.

This landmark paper introduced and demonstrated the effectiveness of **Convolutional Neural Networks (CNNs)** for handwritten digit recognition (MNIST). It is one of the foundational papers of modern deep learning and computer vision, with **tens of thousands of citations**. It popularized the architecture later known as **LeNet-5**.

(May 18, 2026)



Yann LeCun

Chief AI Scientist at Facebook & JT Schwarz Professor at the Courant Institute, [New York University](#)

Verified email at cs.nyu.edu - [Homepage](#)

[AI](#) [machine learning](#) [computer vision](#) [robotics](#) [image compression](#)

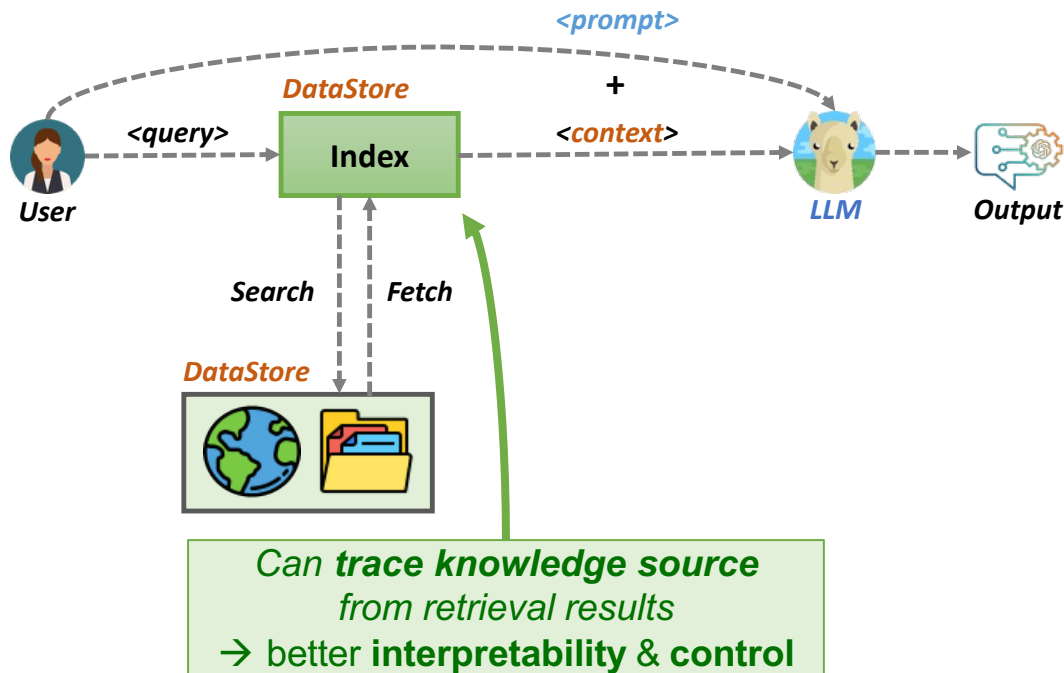


TITLE	CITED BY	YEAR
Deep learning Y LeCun, Y Bengio, G Hinton nature 521 (7553), 436-444	115970	2015
Gradient-based learning applied to document recognition Y LeCun, L Bottou, Y Bengio, P Haffner Proceedings of the IEEE 86 (11), 2278-2324	86568	2002
Backpropagation applied to handwritten zip code recognition Y LeCun, B Boser, JS Denker, D Henderson, RE Howard, W Hubbard, ... Neural computation 1 (4), 541-551	20570	1989

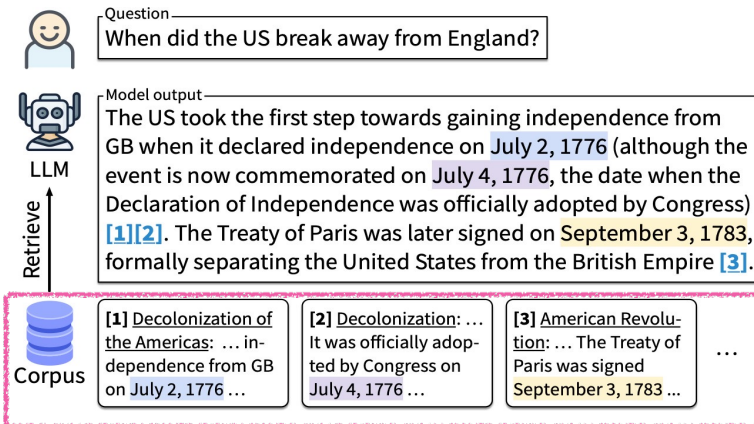
Not even close!

1 Why Retrieval-Augmented LMs?

LLM outputs are hard to interpret and verify



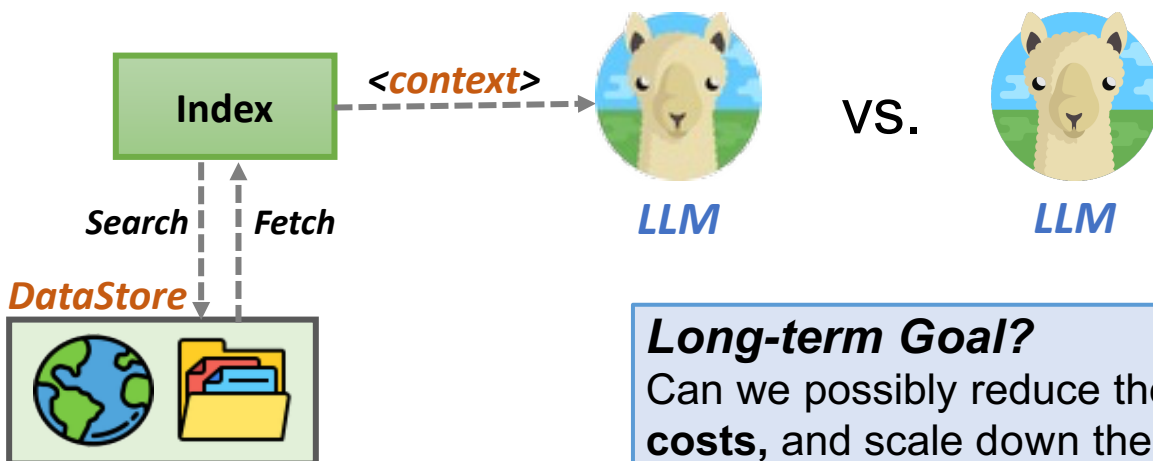
Example - generating text with citations



Enabling Large Language Models to Generate Text with Citations (Gao et al., 2023)

1 Why Retrieval-Augmented LMs?

Can we scale LMs with (test-time) datastore?



Long-term Goal?

Can we possibly reduce the **training** and **inference costs**, and scale down the size of LLMs?



RETRO (Borgeaud et al., 2021)

obtains comparable performance to GPT-3 on
the Pile, despite using 25x fewer parameters

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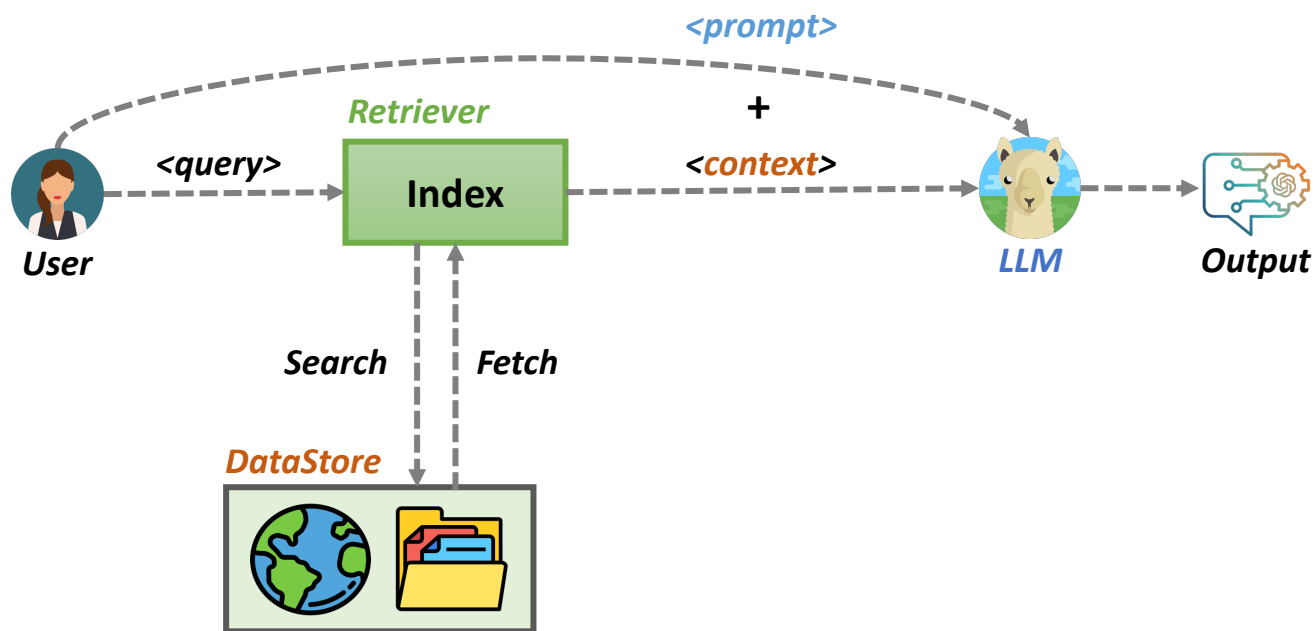
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Reminder: RAG RoadMap



Checklist

Framework/Architecture

- Retriever representation. *Granularity of Datastore?*
- Sizes of LMs vs Datastore?
- Integration of retrieved contexts with LMs.

Learning/Training

- How to train LMs and retriever together?
- Training from scratch? Fine-tuning?
- Pre-training or instruction tuning?

How to Retrieve the Information?

Calculating similarity Score between two pieces of text

Example $\text{sim}(i, j) = \text{tf}_{i,j} \times \log \frac{N}{\text{df}_i}$

$\text{tf}_{i,j}$: # of occurrences of i in j
 N : # of total docs
 df_i : # of docs containing i

Sparse retrieval

Example $\text{sim}(i, j) = \text{Encoder}(i) \cdot \text{Encoder}(j)$

Maps the text into an h -dimensional vector

Dense retrieval

- An entire field of study on how to define or learn these similarity functions better
- There are efficient data structures/infrastructure for supporting fast and accurate search from a large datastore

RETRO (Borgeaud et al., 2021)



Improving language models by retrieving from trillions of tokens

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All authors from DeepMind, [†]Equal contributions, [‡]Equal senior authorship

Framework/Architecture

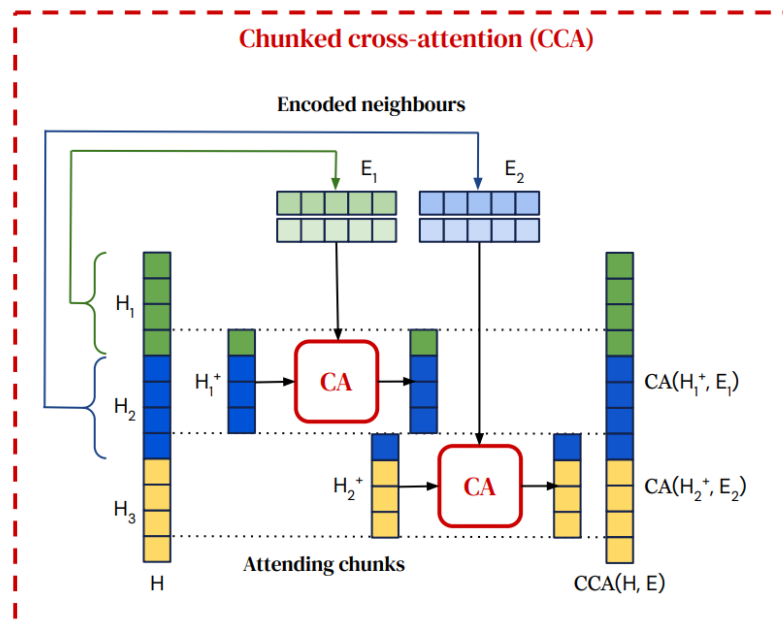
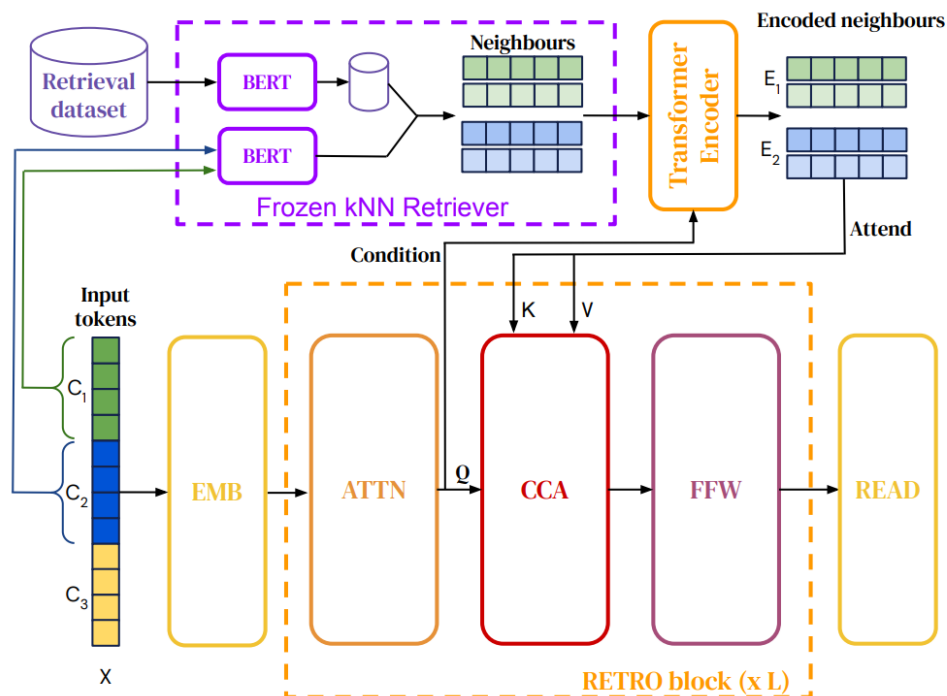
- Retriever representation. *Granularity of Datastore?*
 - **Granularity:** chunks of 64 tokens
 - **Representation:** Frozen BERT encoders
- Sizes of LMs vs Datastore?
 - LMs: 150M-7B parameters
 - Datastore: up to 2T tokens (training tokens 600B tokens)



RETRO (Borgeaud et al., 2021)

Framework/Architecture

- Integration of **retrieved contexts** with **LMs**.
 - The information is integrated into the intermediate layers of Transformers (with an extra encoder)



RETRO (Borgeaud et al., 2021)

Learning/Training

- How to train LMs and retriever together?
- Training from scratch? Fine-tuning?
- Pre-training or instruction tuning?
 - The retriever(“encoder”) is pre-trained and not updated anymore
 - Trained from scratch or fine-tuned (“retrofit”)

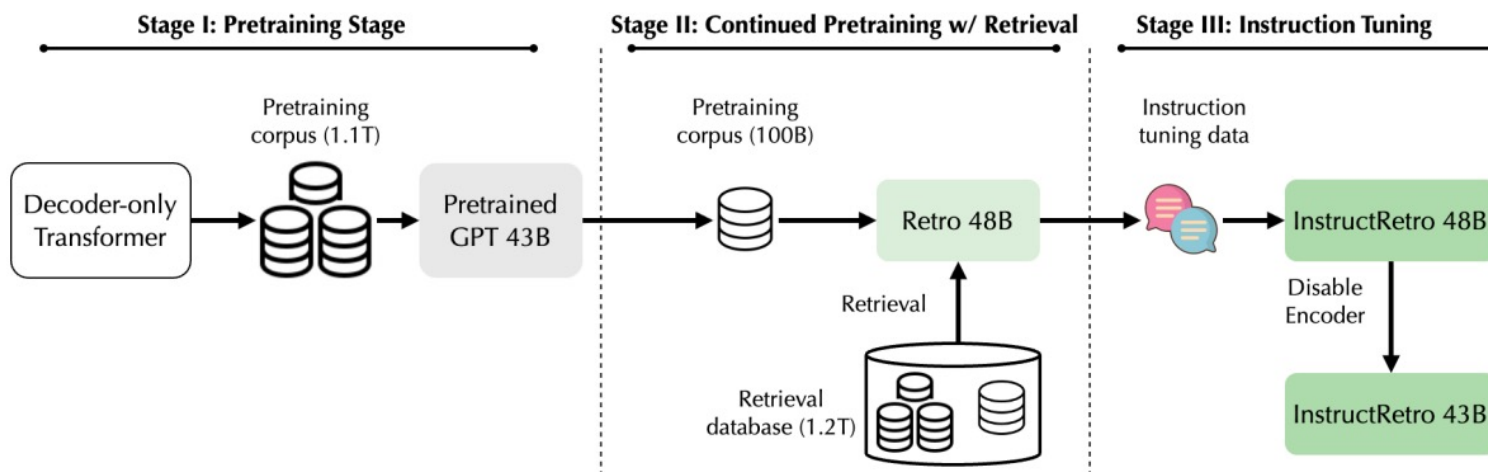


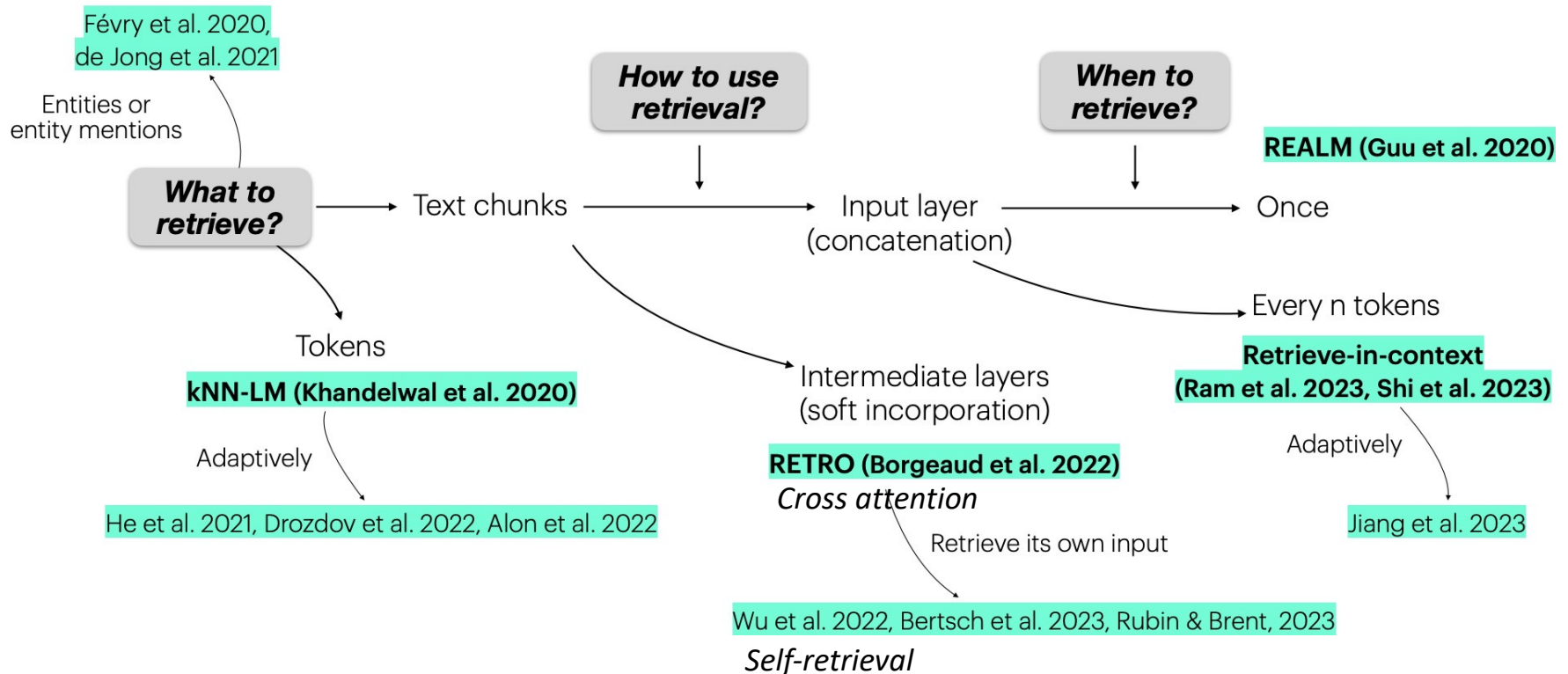
Figure 1. Training pipeline for InstructRetro 48B and InstructRetro 43B.

RETRO (Borgeaud et al., 2021): experiments (Question answering)

Model	Test Accuracy
REALM (Guu et al., 2020)	40.4
DPR (Karpukhin et al., 2020)	41.5
RAG (Lewis et al., 2020)	44.5
EMDR ² (Sachan et al., 2021)	52.5
FID (Izacard and Grave, 2021)	51.4
FID + Distill. (Izacard et al., 2020)	54.7
Baseline 7B (closed book)	30.4
RETRO 7.5B (DPR retrieval)	45.5

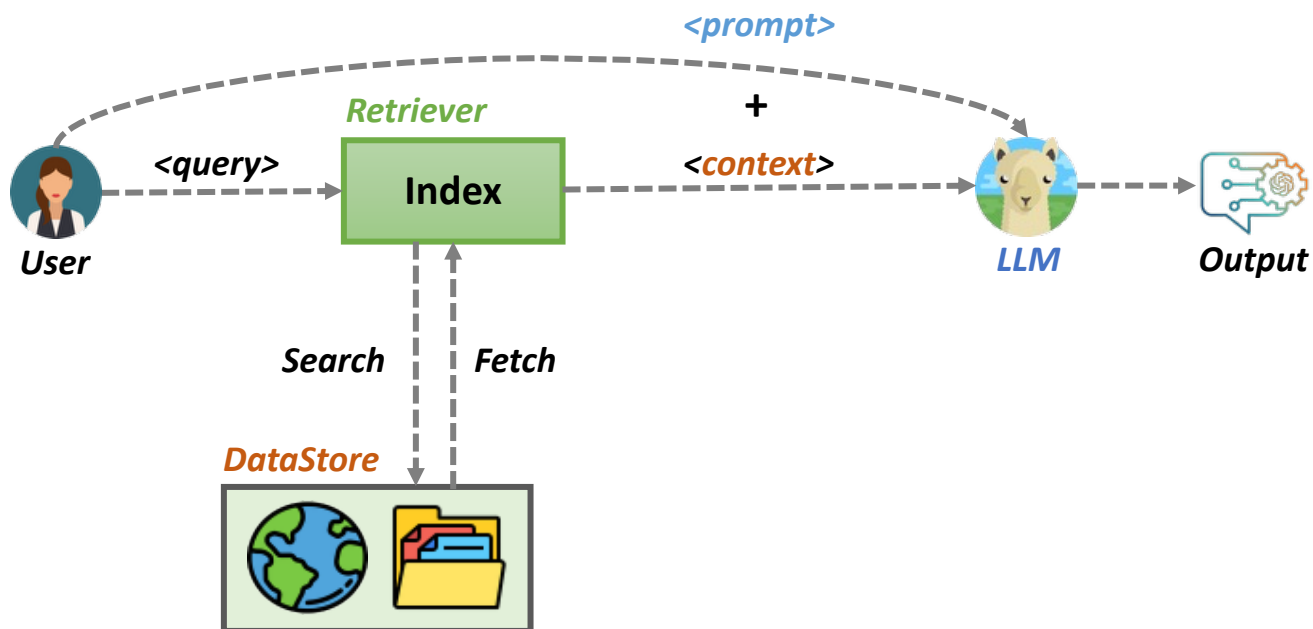
- It is not better than specialised QA models
- No evaluations on any of in-context learnings tasks in GPT-3
- *“With a 2 trillion token database, our Retrieval-Enhanced Transformer (Retro) obtains **comparable performance to GPT-3 on the Pile, despite using 25× fewer parameters.**”*

Other architectures for RALMs



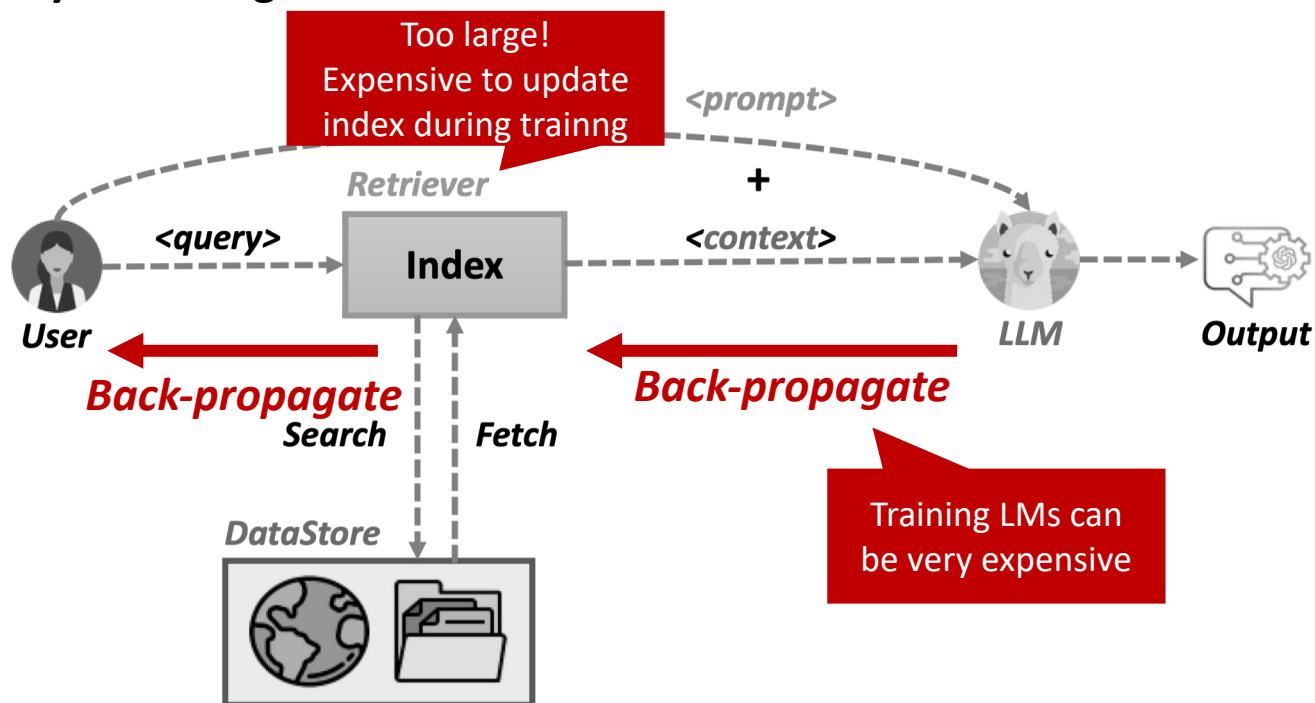
1 RAG framework and Training methods

Why is training so hard for RALMs?



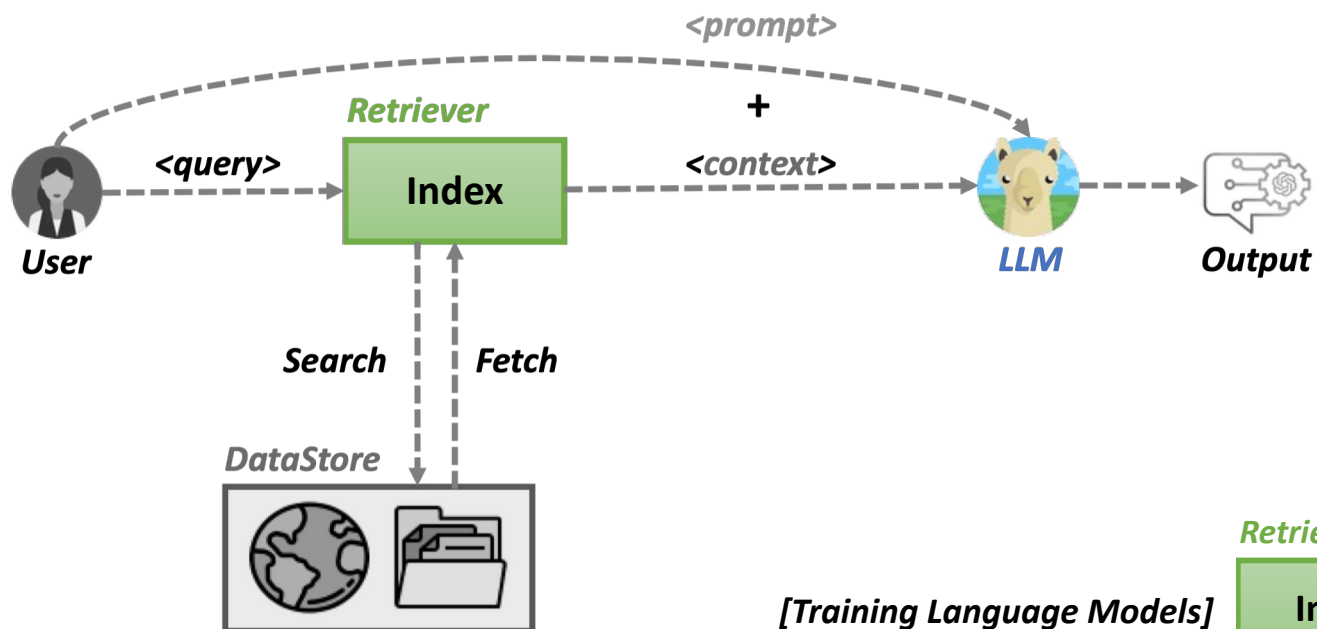
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1 RAG framework and Training methods

Different Training Methods



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2. Knowledge with LLM

- How to Represent the Knowledge?
- Language Models with Knowledge?
- Language Models with Commonsense Knowledge?

LLMs struggle to provide responsible answer based on the static knowledge

** Issue in the specific domain expertise (expert knowledge) required tasks.*

(a) Lack of Responsible Factual Knowledge

LLM + CoT Prompt

Q: Whence is the artist nominated for awards for “Live from Paris”?

A: Let’s think step by step.

As of my last knowledge update in January 2022, I don't have specific information about an artist nominated for awards ...



Where can we get those “Knowledge”?

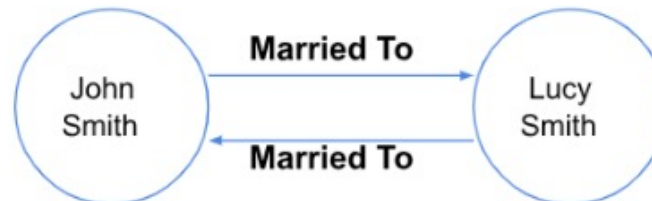
A knowledge graph?

A data structure representing information as a network of entities and their relationships. It is a directed labeled graph in which domain-specific meanings are linked with nodes and edges (relationships).

Table Representation

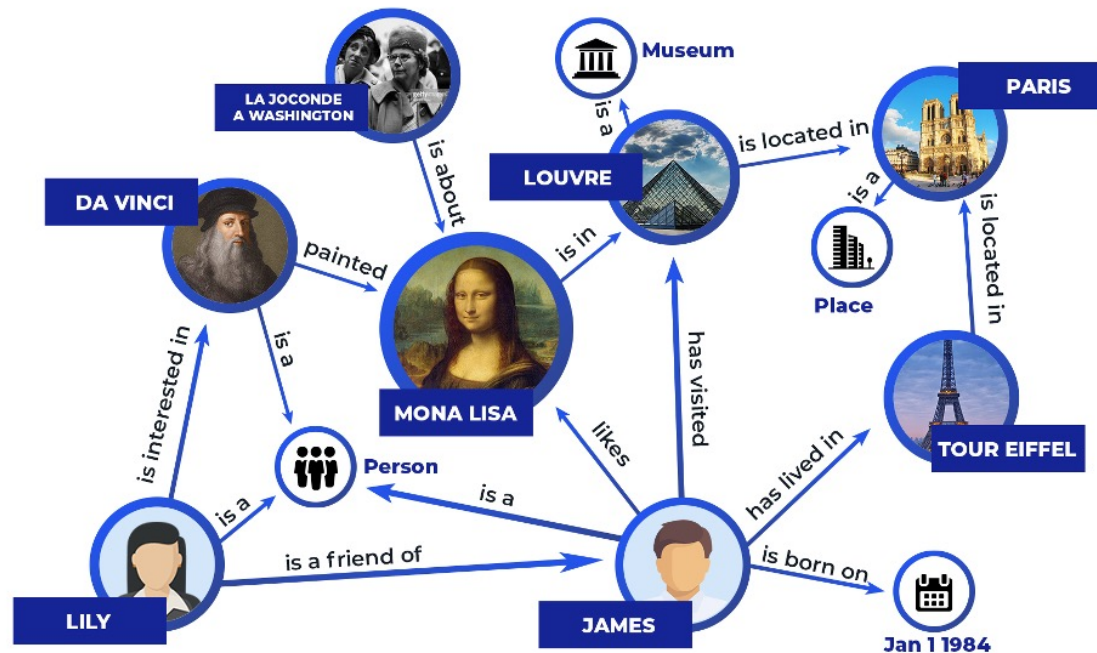
Name	Married to
John Smith	Lucy Smith
Lucy Smith	John Smith

Graph Representation



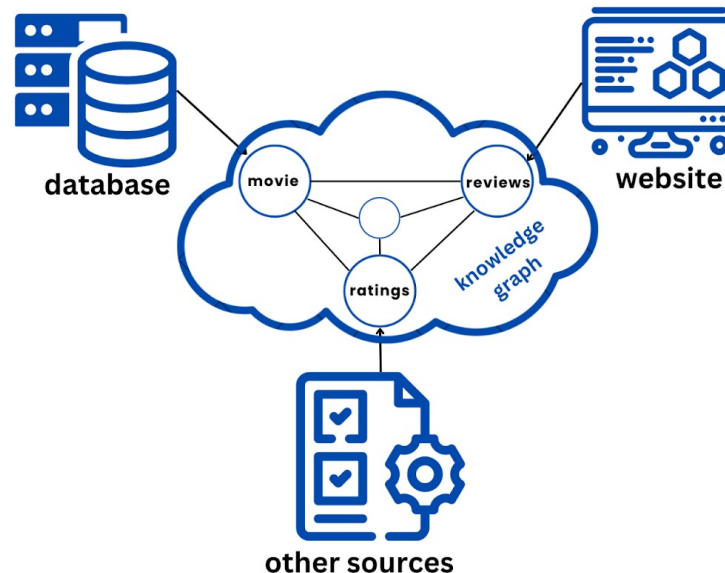
2 Knowledge Graph? Knowledge Base?

A knowledge graph?



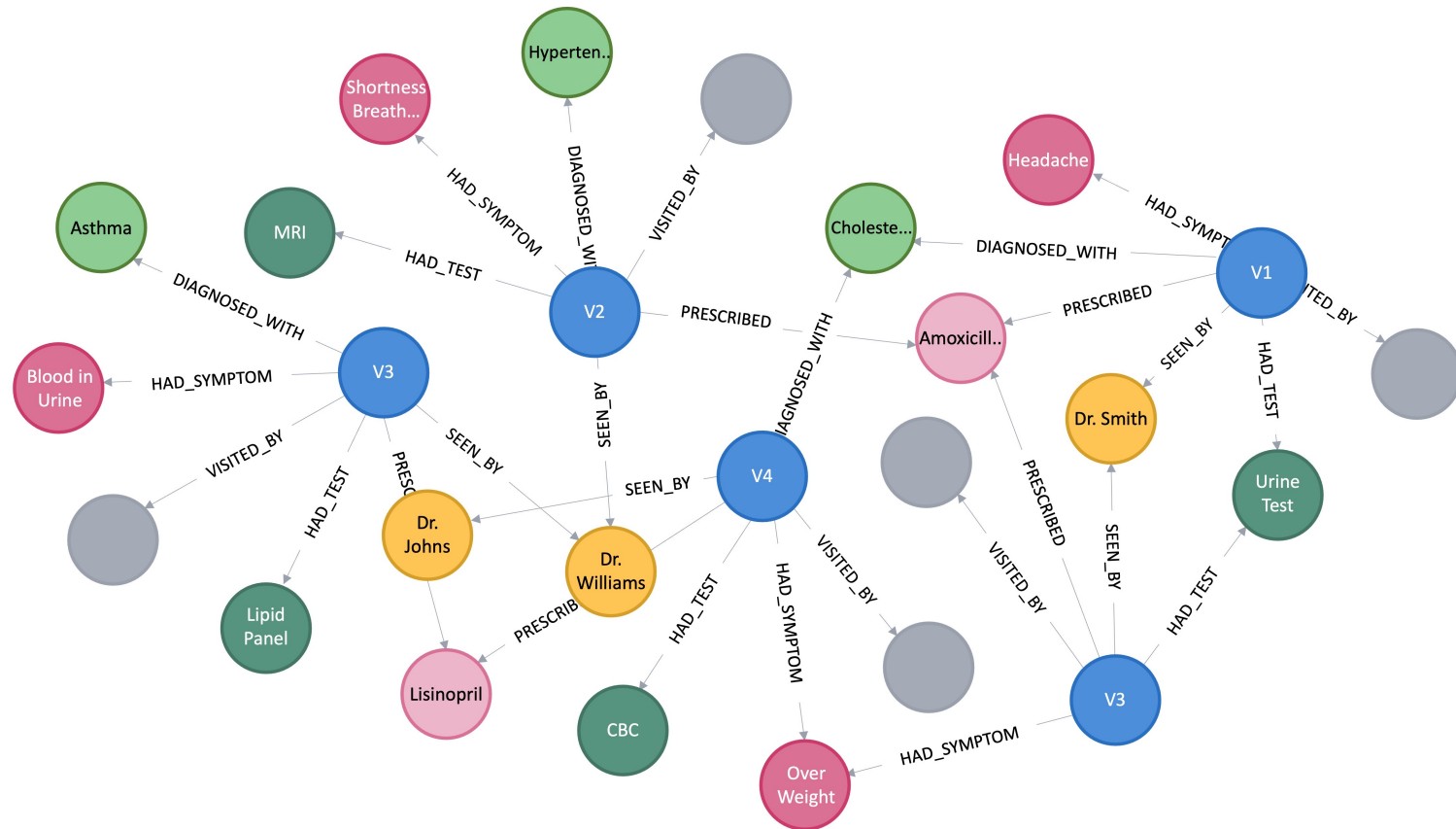
Ability to Integrate Data from Multiple Sources

Combining data from numerous sources can be complex. When we bring together structured data (like databases), semi-structured data (like XML files), and unstructured data (like text), it can get messy. Thankfully, knowledge graphs are highly flexible and excel at integrating these pieces into one clear picture.



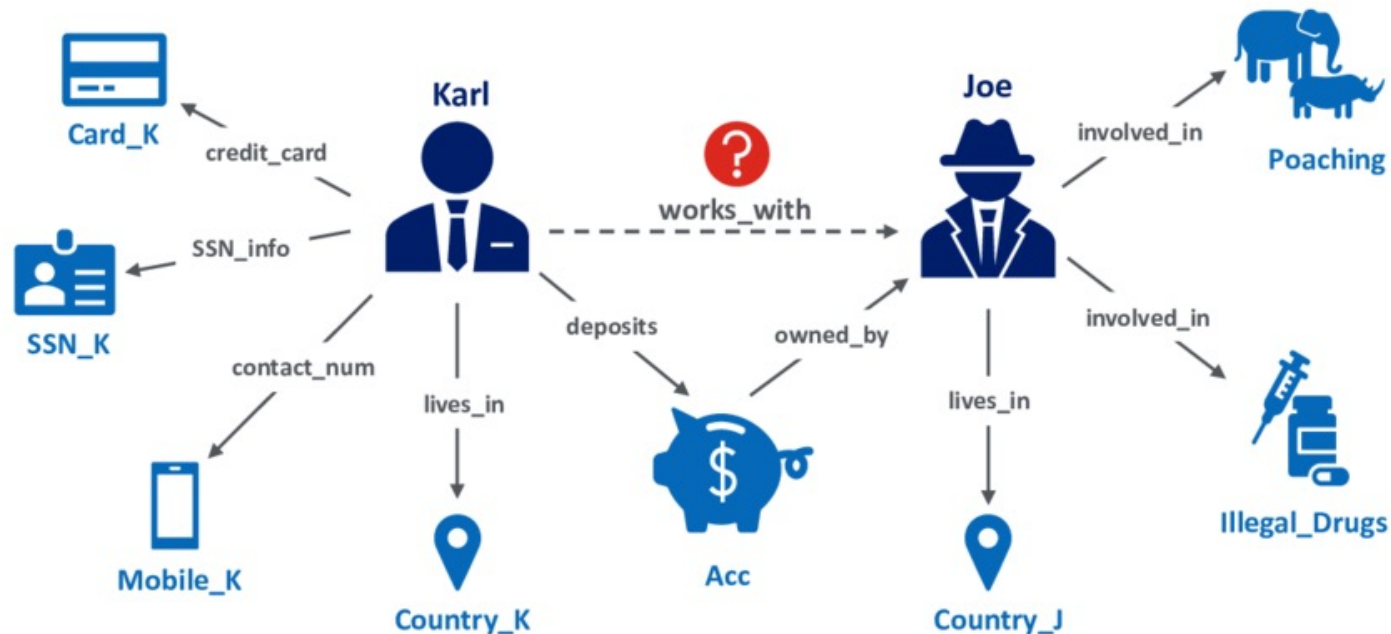
2 Knowledge Graph? Knowledge Base?

Domain Specific KG: Clinical Knowledge Graph



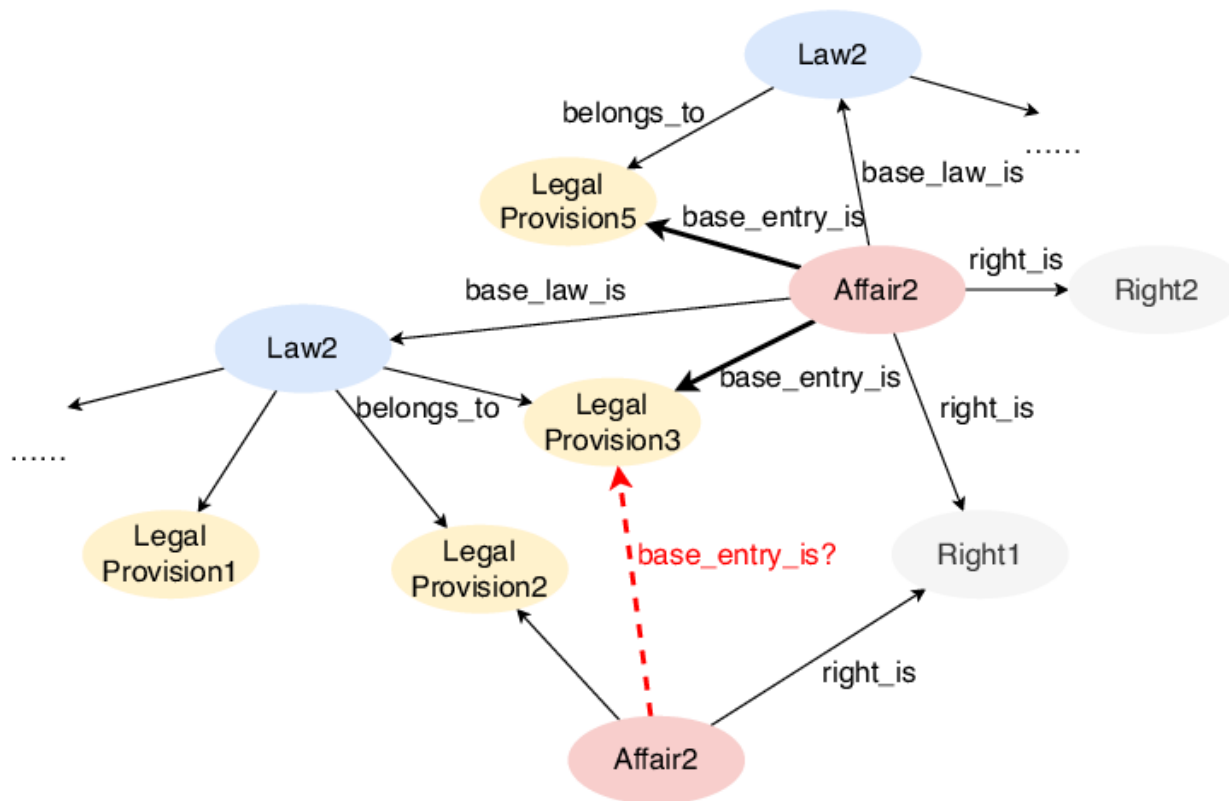
2 Knowledge Graph? Knowledge Base?

Domain Specific KG: Financial Knowledge Graph



2 Knowledge Graph? Knowledge Base?

Domain Specific KG: Legal Knowledge Graph



How to adapt Knowledge to the Language Models?

(a) Lack of Responsible Factual Knowledge

LLM + CoT Prompt

Q: Whence is the artist nominated for awards for “Live from Paris”?

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(b) Cognition Gap with Knowledge Retrievers

LLM + Knowledge Retriever

Q: Whence is the artist nominated for awards for “Live from Paris”?

K: (Sakira, *music.artist.album*, Live from Paris)
 (Shakira, *music.artist.album*, Las de la Intuicion)
 (Shakira, *people.person.places_lived*, Los Angeles) ...

Los Angeles

The high relevance of fragmented knowledge facts doesn't necessarily imply the usefulness for LLM reasoning

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Los Angeles

(c) KG-CoT (Ours)

LLM + Chain-of-Thought Prompting over Knowledge Graphs

Q: Whence is the artist nominated for awards for “Live from Paris”?

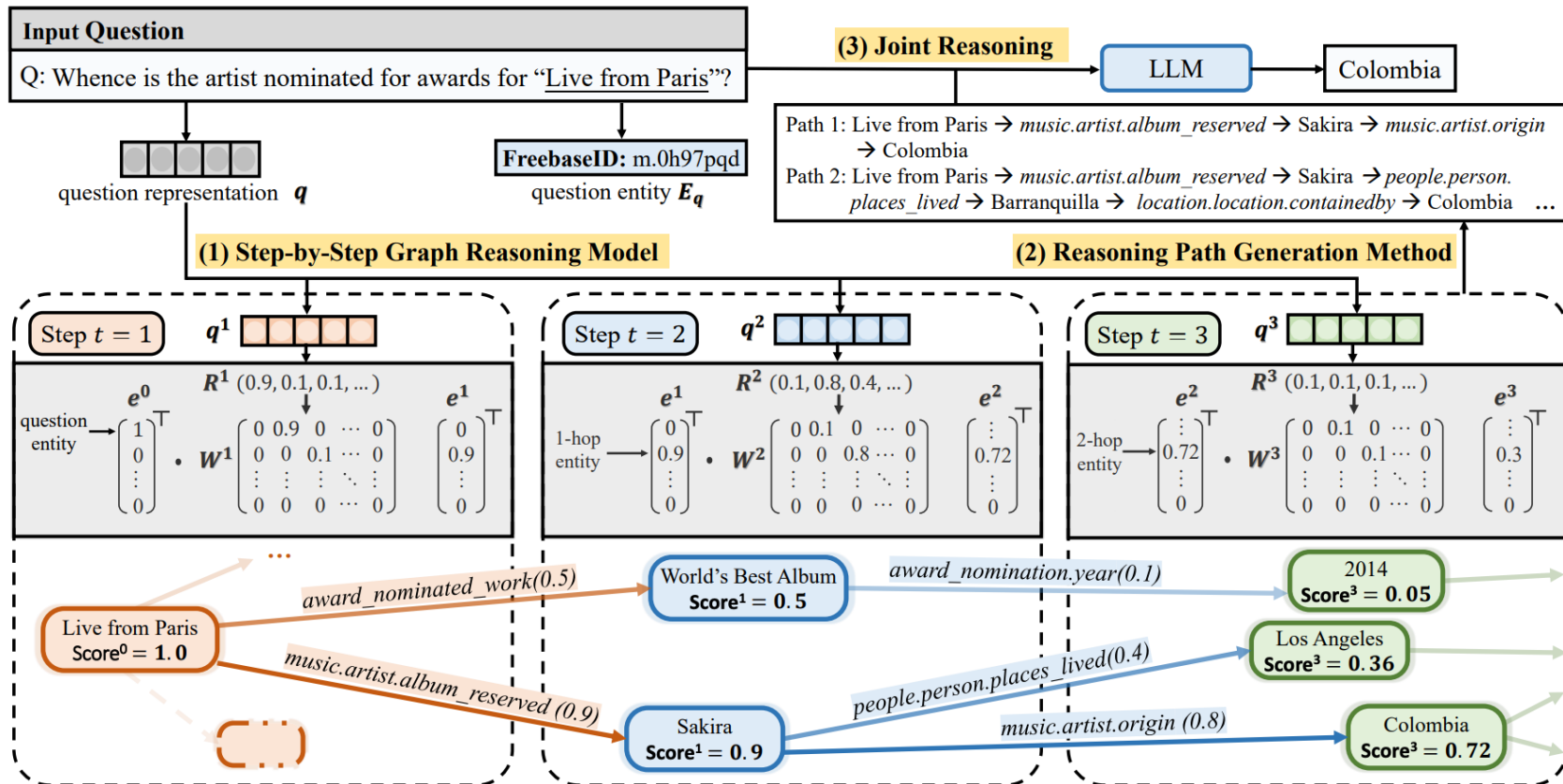
K: Path1: Live from Paris → *music.artist.album_reversed* → Sakira
 → *music.artist.origin* → Colombia ...

Based on the reasoning paths, we can infer that Shakira, the artist behind the album “Live from Paris”, hails from Colombia.

The high relevance of fragmented knowledge facts doesn't necessarily imply the usefulness for LLM reasoning

Maybe something else... 🤔

How to adapt Knowledge to the Language Models? (KG-COT)



How to adapt Knowledge to the Language Models?

Method	KB	WebQSP	CWQ
<i>Baselines</i>			
GFC [Xie <i>et al.</i> , 2022]	✓	76.8	50.4
UniKGQA [Jiang <i>et al.</i> , 2023]	✓	76.6	52.2
ChatGPT+ToG-R [Sun <i>et al.</i> , 2023]	✓	75.8	58.9
GPT-4+ToG-R [Sun <i>et al.</i> , 2023]	✓	81.9	69.5
<i>Llama2-7B</i>			
CoT	×	46.1	27.6
KG-CoT	✓	72.4	46.7
Gain		(+26.3)	(+19.1)
<i>Llama2-13B</i>			
CoT	×	47.2	29.9
KG-CoT	✓	74.6	50.0
Gain		(+27.4)	(+20.1)
<i>ChatGPT</i>			
CoT	×	62.2	38.8
KG-CoT	✓	82.1	51.6
Gain		(+19.9)	(+12.8)
<i>GPT-4</i>			
CoT	×	67.3	46.0
KG-CoT	✓	84.9	62.3
Gain		(+17.6)	(+16.3)

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LLM with Commonsense Knowledge

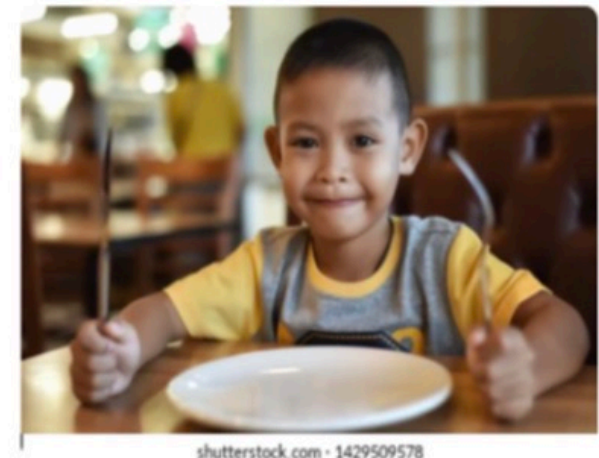
LLaVA Chatbot



what are they doing?

In the image, a group of people is gathered around a dining table, which is set for a meal. They appear to be in the middle of a festive occasion, possibly a holiday dinner, given the decorations and the presence of wine glasses. One person is standing and holding a red object, which could be a napkin or a piece of cloth, and seems to be making a gesture or a toast. The atmosphere looks cheerful and convivial, with everyone engaged in conversation and enjoying the meal.

(A) Example with LLaVA (01.06.2024)



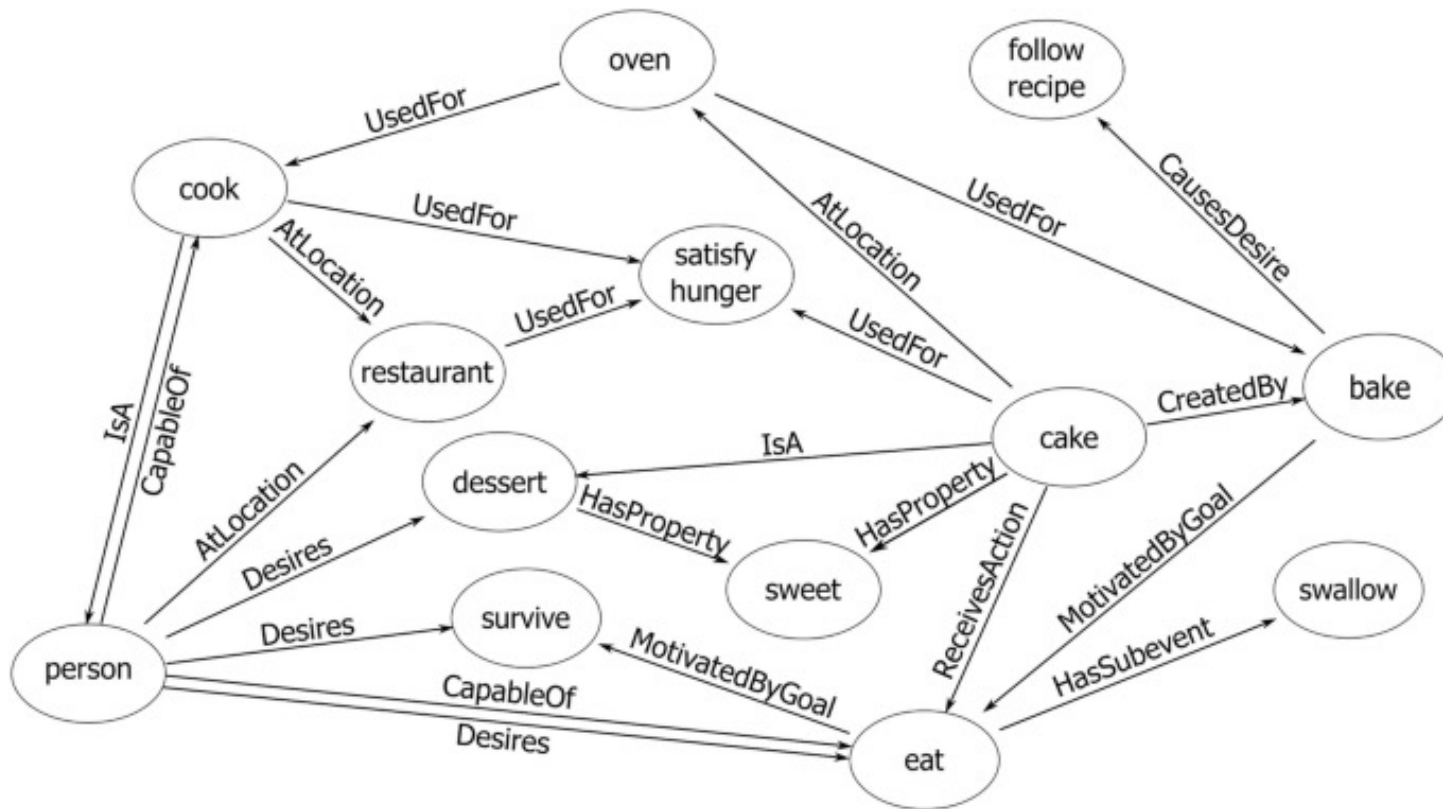
What is he doing?



The boy in the image appears to have finished his meal at a restaurant. He sits at a table with an empty plate in front of him, holding utensils in both hands. His facial expression suggests he is happy and possibly excited with the food that he has.

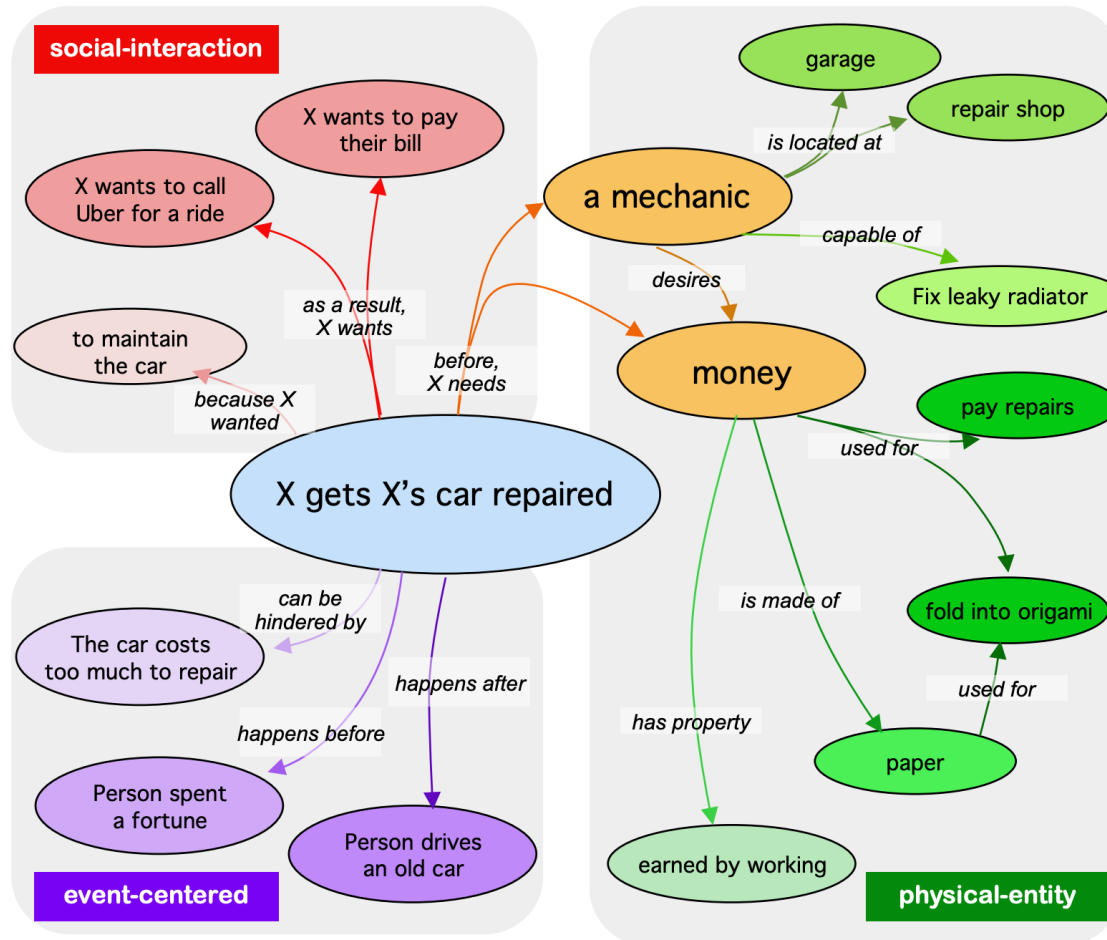
(B) Example with ChatGPT4o (01.06.2024)

ConceptNet (Semantic Networks) – Speer, 2016



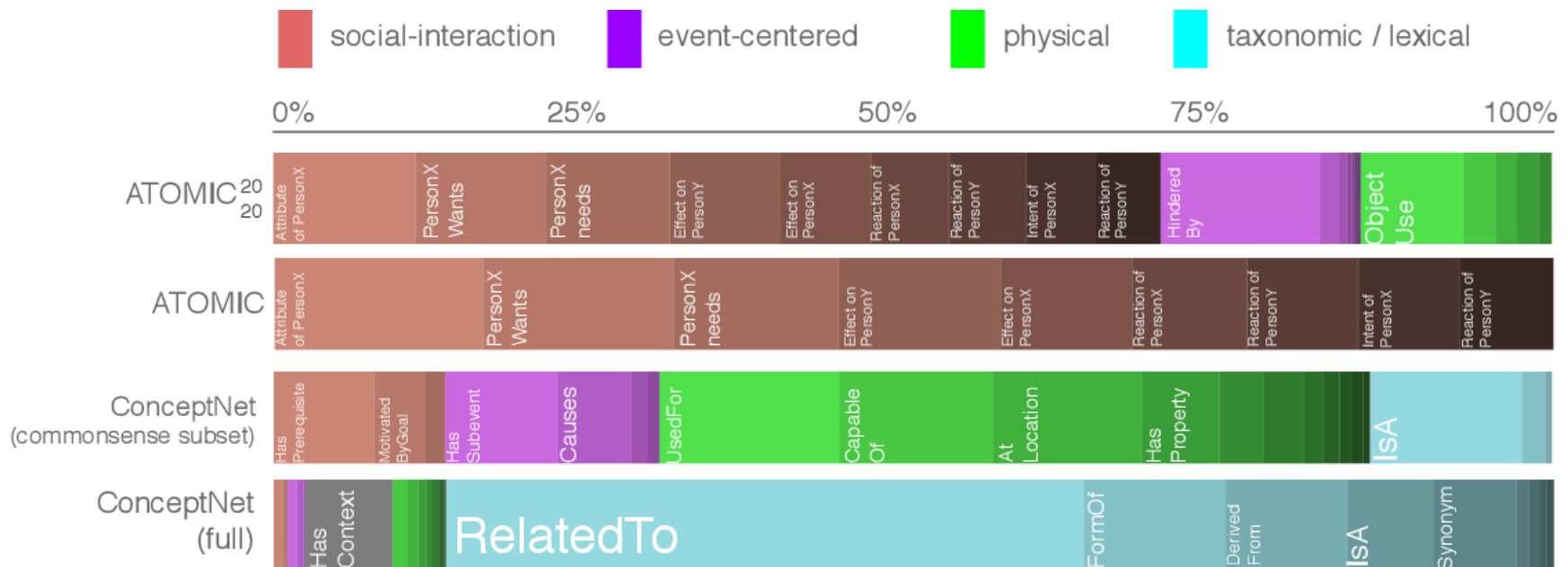
ATOMIC 2020:

On Symbolic and Neural Commonsense Knowledge Graphs - Hwang 2020



2 LLM with Commonsense Knowledge

ATOMIC 2020: On Symbolic and Neural Commonsense Knowledge Graphs - Hwang 2020

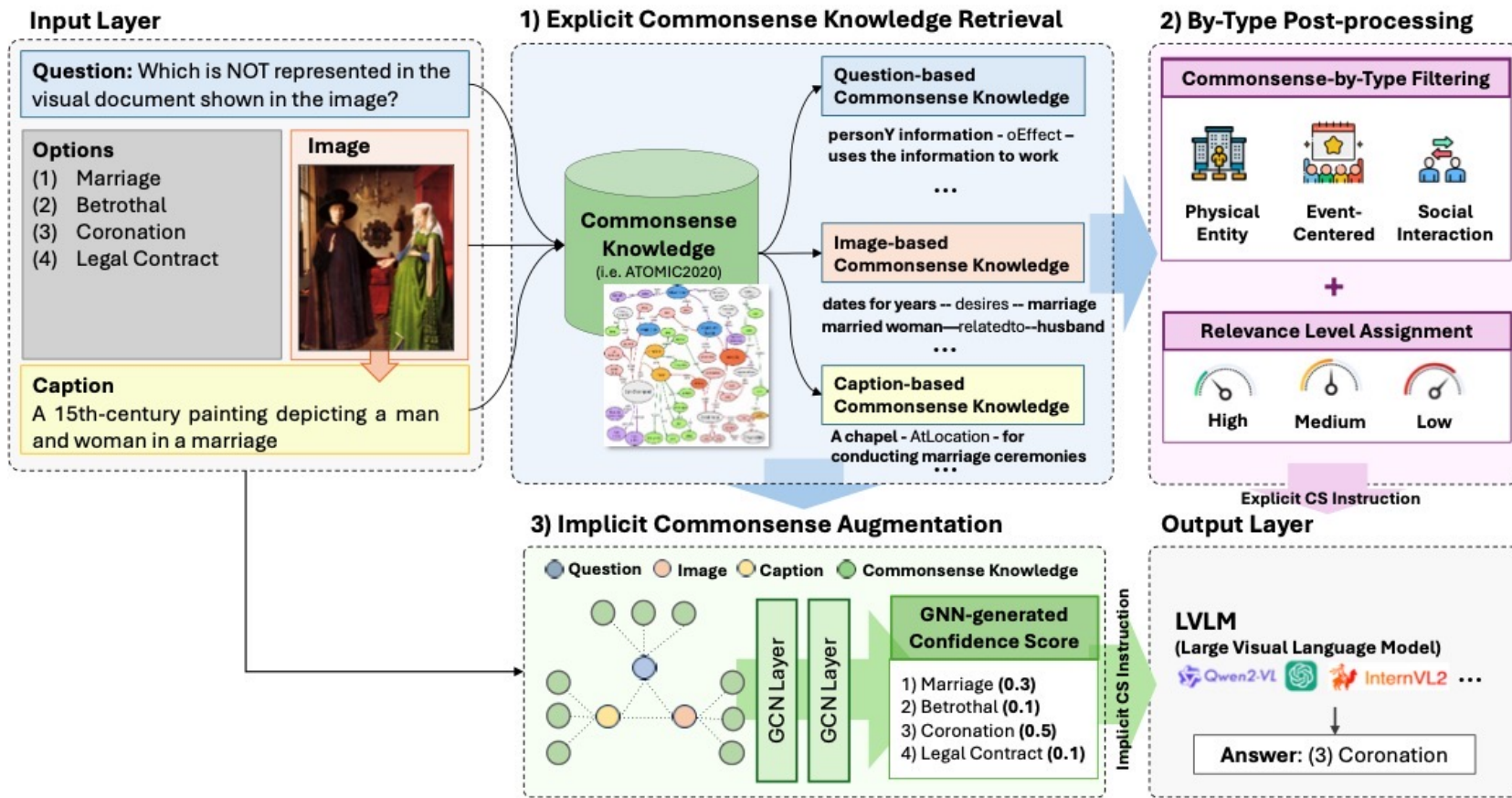


ATOMIC 2020:

On Symbolic and Neural Commonsense Knowledge Graphs - Hwang 2020

Relation	Transformed Format
<i>Physical-Entity</i>	
ObjectUse	is used for
AtLocation	is at
MadeUpOf	is made up of
HasProperty	can be
CapableOf	is capable of
Desires	desires
NotDesires	does not desire
<i>Event-Centered</i>	
IsAfter	occurs after
HasSubEvent	has sub-event
IsBefore	occurs before
HinderedBy	is hindered by
Causes	causes
xReason	is because someone
isFilledBy	is filled by
<i>Social-Interaction</i>	
xNeed	then someone needs
xAttr	then someone has attributes
xEffect	then someone has the effect
xReact	then someone reacts with
xWant	then someone wants
xIntent	then someone intends
oEffect	then the effect on another is
oReact	then another reacts with
oWant	then another one wants

Commonsense Knowledge Adapted Visual Question Answering (Yang, 2025)



How does it look like?

Background

You are an advanced Vision-Language Model assistant designed to answer multiple-choice questions based on a given image. Your task is to select the most appropriate option from the provided answer choices. You are given an input image, a question related to the image, the image caption, multiple-choice answer options, and both explicit and implicit commonsense knowledge.

Explicit commonsense knowledge consists of statements related to the input, categorized as image-related commonsense, question-related commonsense, and caption-related commonsense. Implicit commonsense knowledge includes the relevance level (e.g., highly relevant, relevant, less relevant) assigned to each explicit commonsense statement and the confidence of each candidate option, where higher values indicate a greater likelihood of being correct.

Your objective is to integrate the explicit and implicit commonsense knowledge with the provided information to generate a step-by-step reasoning. Based on this rationale, you will select the most appropriate answer from the given options.

Input Information

Image:



Question: Which of these oceans does the prime meridian intersect?

Caption: An image of a world map with labeled continents and oceans.

Options:

- A. "the Atlantic Ocean"
- B. "the Indian Ocean"
- C. "the Pacific Ocean"

Explicit Commonsense Knowledge

Image-Related Commonsense:

- The Atlantic Ocean is at the western hemisphere. (Highly Relevant)
- A world traveler is capable of crossing many time zones. (Relevant)

Question-Related Commonsense:

- A traveler is capable of crossing geographical borders. (Highly Relevant)
- Someone who is far from home might want to measure the distance. (Less Relevant)

Caption-Related Commonsense:

- The Atlantic Ocean is used for separating continents. (Highly Relevant)
- If someone sees the ocean, they might think of traveling to it. (Relevant)

Implicit Commonsense Knowledge (Confidence for Each Option)

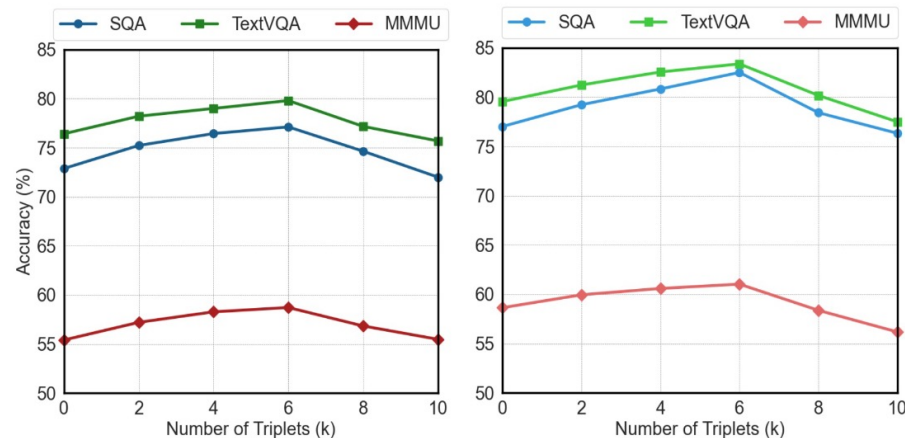
- A: 0.6
- B: 0.05
- C: 0.35

Commonsense Knowledge Adapted Visual Question Answering

Models	None	CS Sources			CS Categories			All CS
		CS-Q	CS-I	CS-C	CS-PE	CS-EC	CS-SI	
ScienceQA _{IMG}								
LLaVA1.6	67.50	68.83	<u>71.56</u>	70.35	<u>71.12</u>	69.01	70.83	72.30
BLIP3	70.00	71.56	<u>73.88</u>	72.97	<u>74.03</u>	71.57	71.05	74.30
InternVL2	71.99	72.58	<u>74.37</u>	73.91	<u>74.56</u>	73.09	73.21	74.62
Qwen2VL	71.39	72.21	<u>74.83</u>	71.86	<u>74.22</u>	72.03	72.57	75.95
GPT4o	76.45	77.34	<u>79.83</u>	77.17	<u>79.63</u>	77.52	78.87	81.22
TextVQA _{val}								
LLaVA1.6	62.30	63.55	<u>64.82</u>	64.23	64.77	<u>65.05</u>	64.89	65.20
BLIP3	67.80	68.49	<u>69.64</u>	68.29	69.12	<u>69.64</u>	69.24	69.80
InternVL2	73.21	74.06	<u>75.19</u>	74.81	74.60	<u>75.01</u>	74.82	75.30
Qwen2VL	75.30	76.07	<u>77.63</u>	77.05	76.57	<u>78.02</u>	76.85	78.90
GPT4o	78.98	79.34	<u>81.25</u>	80.63	80.93	<u>81.51</u>	81.22	82.13
MMMU _{val}								
LLaVA1.6	48.38	49.27	<u>53.52</u>	49.85	52.03	52.57	<u>53.10</u>	54.30
BLIP3	41.31	42.54	<u>45.89</u>	42.19	44.12	<u>46.03</u>	45.89	47.60
InternVL2	51.00	52.17	<u>55.48</u>	54.21	<u>54.23</u>	52.67	53.50	55.80
Qwen2VL	51.10	52.69	<u>55.89</u>	54.83	53.60	<u>54.57</u>	54.10	57.42
GPT4o	55.89	56.53	<u>58.79</u>	56.21	<u>58.12</u>	57.57	57.89	60.87

None: without commonsense knowledge

Commonsense Knowledge Adapted Visual Question Answering

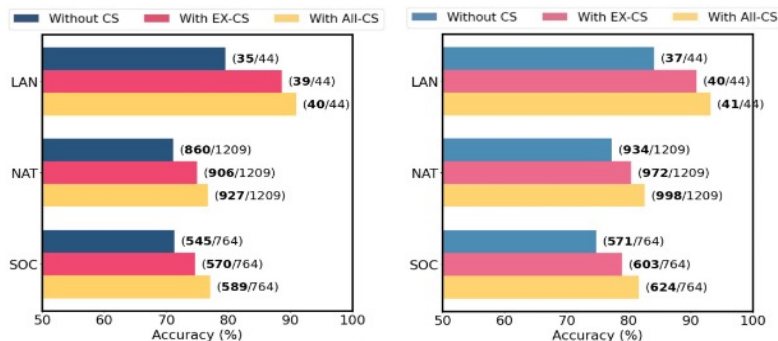


(a) Qwen2VL

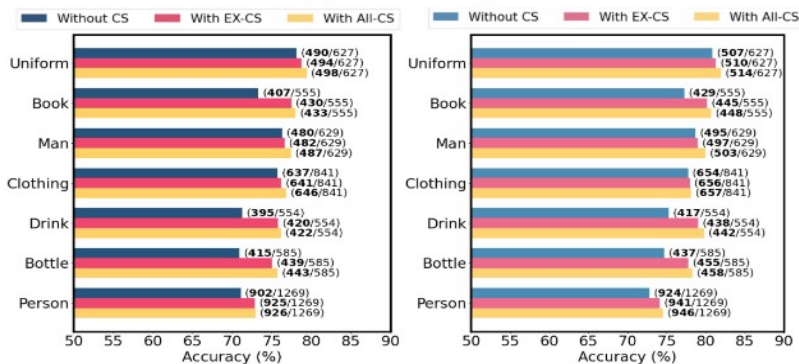
(b) GPT4o

Figure 4: Effect of number of knowledge triplets k : Comparison between Qwen2VL and GPT4o.

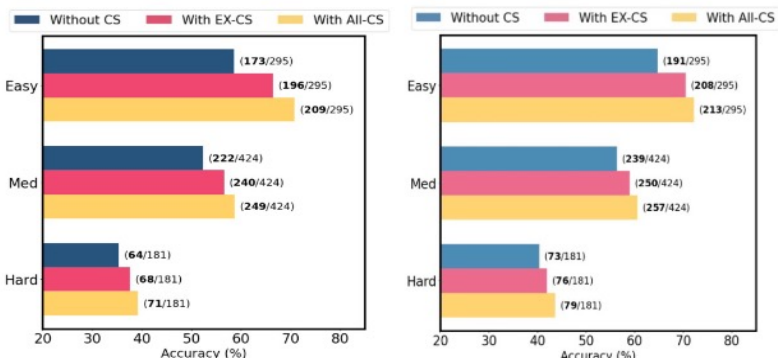
2 LLM with Commonsense Knowledge



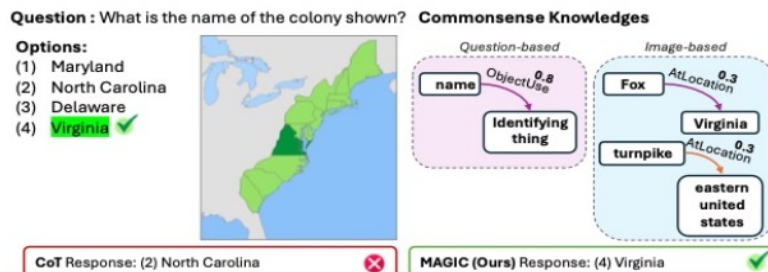
(a) ScienceQA



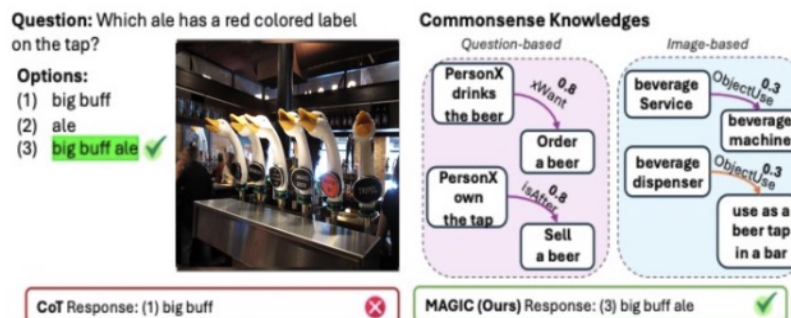
(b) TextVQA



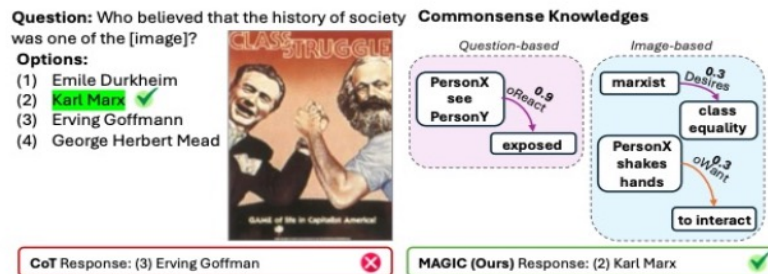
(c) MMMU



(a) A case from ScienceQA dataset



(b) A case from TextVQA dataset



(c) A case from MMMU dataset